Structure of Chromosomes, Cell Cycle and Cell Division

CHAPTER: 1

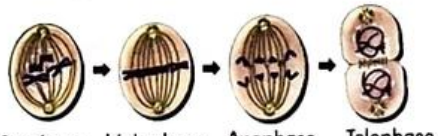
WHY DO CELLS DIVIDE

- Cells divide for growth, development, repair of worn-out tissues and reproduction.
- To facilitate the exchange of materials.
- To control DNA overloading.

Cell Division

Mitosis-Equation division

- Production of diploid cells
- For growth and development



Meiosis-Reductional division

- Production of haploid cells
- For gamete (sperm or egg) formation

Meiosis I

- Prophase I
- Metaphase I
- Anaphase I
- Telophase I

Meiosis II

- Prophase II
- Metaphase II
- Anaphase II
- Telophase II

Cell cycle is a series of events that take place in cell leading to the duplication of its DNA and the subsequent division of the cell to produce daughter cells.

Cell Cycle

Interphase

- G1
- S
- G2

M phase

Karyokinesis

- Prophase
- Metaphase
- Anaphase
- Telophase

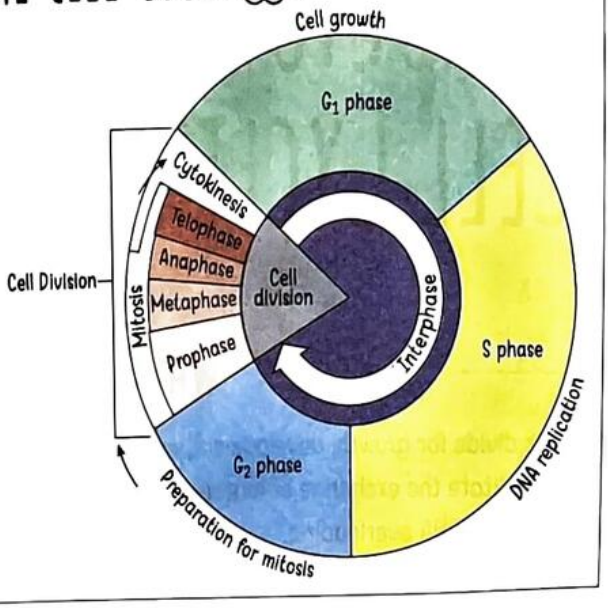
Cytokinesis

In Animal Cell
Starts with appearence of a furrow in the plasma membrane.

In Plant Cell
Cell wall formation starts in the centre.

CONTROL OF THE CELL CYCLE

- Regulatory proteins called **cyclins** control the cell cycle at checkpoints:
- **G1 Checkpoint**-decides whether or not cell will divide.
- **S Checkpoint**-determines if DNA has been properly replicated.
- **Mitotic Spindle Checkpoint**-ensures chromosomes are aligned at mitotic plate.



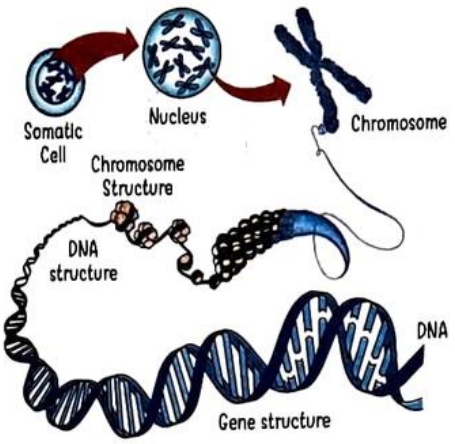
CHROMATIN AND CHROMATID

Chromatin refers to a substance found in the cell nucleus that's composed primarily of DNA and proteins. When cells divide, chromatin condenses to form chromosomes which split into two identical strands called chromatids.

GENE STRUCTURE OF DNA

Deoxyribonucleic Acid

- DNA is a group of molecules that responsible for carrying and transmitting the hereditary materials from parents to offspring.
- The basic building blocks of DNA are nucleotides, which are composed of a sugar group, a phosphate group, and a 4 types of nitrogenous bases.
- Adenine (A), Thymine (T), Guanine (G) and Cytosine (C) are four types of nitrogen bases present in DNA.



Chromosomes

Rod-shaped or thread like DNA-containing bodies of a cell nucleus that contain all or most of the genes of an organism and can be seen especially during cell division.

- Autosomes (Non-body Chromosomes)
- Heterosomes or allosomes (Sex - Chromosomes) (X and Y Chr.)

GENETICS

CHAPTER: 2

GENETICS

The branch of biology focused on the study of heredity and variations in organisms.

- The phenomenon of passing the characters from parents to progeny through successive generations is called **heredity.** The fundamental units of heredity are genes.
- **Genes** are the specific parts of a chromosome (DNA segment) that serves as a unit of hereditary information and codes for a specific protein or function.

Mendelian Genetics

Gregor Johann Mendel (1866); known as the **'Father of Genetics'** gave the fundamental concepts of genetics based upon his extensive research on **pea** plants (***Pisum sativum***).

Characteristics of pea plants Gregor Mendel used in his inheritance experiments

Seeds		Flower colour	Pod		Position of inflorescence	Stem size
Shape	Cotyledons		Shape	Colour		
round	yellow	white	inflated	yellow	axial	tall
wrinkled	green	violet-red	constricted between the seeds	green	terminal	dwarf

MENDEL'S LAWS OF INHERITANCE

Law of Dominance: This law states that in a hybrid, the dominant allele expresses itself while the recessive allele is suppressed.

Law of Segregation: It states that the two alleles for a character segregate or separate during gamete formation.

Law of Independent Assortment: This law explains that the factors for different pairs of contrasting characters behave independently of each other during gamete formation.

SOME BASIC TERMS RELATED TO GENETICS

- **Monohybrid Cross:** This refers to a genetic cross involving a single trait with two alleles. Typically, in a classic Mendelian monohybrid cross, the phenotype ratio is 3:1 (dominant to recessive traits) and the genotype ratio is 1:2:1 (homozygous dominant, heterozygous, homozygous recessive).
- **Dihybrid Cross:** A dihybrid cross involves two pairs of different traits, each with two alleles. The phenotype ratio is usually 9:3:3:1 in a Mendelian dihybrid cross.

- **Heterozygous:** An individual having two different alleles for a specific gene (e.g., Aa).
- **Homozygous:** An individual having two identical alleles for a particular gene (e.g., AA or aa).
- **Allele:** One of two or more alternative forms of a gene, result in different traits, even though they are for the same gene.
- **Dominant:** An allele is dominant if it is expressed in the phenotype of an organism that has only one copy of that allele.
- **Recessive:** An allele is recessive if it is expressed in the phenotype only when two copies are present (i.e., not expressed when a dominant allele is present).
- **Phenotype:** The set of observable characteristics of an organism resulting from the interaction of its genotype with the environment.
- **Genotype:** The genetic constitution of an organism, determining its specific traits.

CROSSING OVER

Referring to the process where homologous chromosomes exchange segments of their genetic material during meiosis, results in genetic recombination, which is important for genetic diversity.

- It ensures that offspring inherit a mix of genes from both parents, contributing to variations in traits among individuals.
- "Variations" in genetics denote the differences in physical or genetic characteristics among individuals within a population.
- These variations are key to the diversity observed within species and are essential for the process of natural selection and evolution.

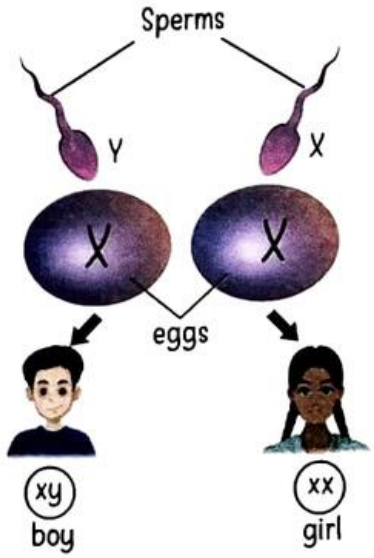
MUTATION

A change in the DNA sequence of an organism's genome.

- Can occur due to various factors such as errors during DNA replication, exposure to certain chemicals or radiation, and can result in genetic diversity.
- Mutations can be beneficial, harmful, or neutral in their effects on the organism.
- They play a crucial role in evolution by introducing new genetic variations, and can also be the cause of certain genetic disorders or diseases.

SEX DETERMINATTION IN HUMANS

The sex of a child depends on only male parent, whether the sperm carrying an X or Y chromosome fertilises the egg.



SEX-LINKED INHERITANCE DISEASES

Some a diseases are linked to sex chromosomes, such as hemophilia and colour blindness. These are more common in males than females due to the presence of a single X chromosome in males.

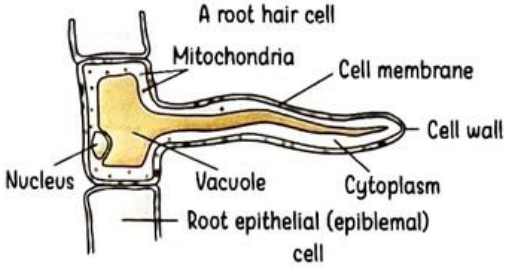
ABSORPTION BY ROOTS

CHAPTER: 3

Water is a universal solvent essential for all the important life processes. Land plants absorb water from the soil through root hairs which are unicellular extensions of roots.

ABSORPTION BY ROOTS

Roots have a large number of root hairs as it increases the overall surface area and hence, allows an increase in the absorption of water and dissolved minerals from the soil during the process of osmosis.



Absorption of Water and Mineral Salts

- Osmosis
  - Exosmosis
  - Endosmosis
- Diffusion
- Imbibition
- Active Absorption
- Passive Absorption

ACTIVE AND PASSIVE TRANSPORT

These are processes that cells use to move substances across their membranes. Active transport requires energy to move substances against their concentration gradient, while passive transport does not require energy and moves substances along their concentration gradient.

TURGIDITY AND FLACCIDITY

Turgidity refers to a state where cells are swollen with water, making the plant rigid and upright. Flaccidity, on the other hand, is when cells lose water and become limp, causing the plant to wilt.

ROOT PRESSURE

This is a pressure developed in the roots of plants, which helps in pushing water and nutrients upwards from the soil into the plants. It is caused by water entering the roots from the soil, creating a pressure that helps drive the water upward.

PLASMOLYSIS AND DEPLASMOLYSIS

Plasmolysis is the process where plant cells lose water in a hypertonic solution, leading to the cell membrane detaching from the cell wall. Deplasmolysis is the reverse process, where cells reabsorb water, returning to their normal state after being plasmolysed.

OSMOTIC PRESSURE

This is the pressure required to prevent the movement of water across a selectively permeable membrane due to osmosis.

ASCENT OF SAP

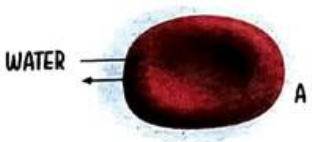
The sap is the fluid that moves through the plant, consisting of water with dissolved minerals and sugars. This upward movement of liquid is known as the ascent of sap. This movement of water occurs due to attraction between the water molecules **(cohesion)** and attraction of the water molecules to the walls of the xylem tracheids and vessels **(adhesion).** The Force responsible for the ascent of sap is the Capillary force, Root pressure, Transpiration pull.

Transpiration pull: The force that draws water upward from the roots to the leaves due to water evaporation from the stomata.

Fate of an animal cell and a plant cell when placed in different kinds of salt-solution

- Isotonic Solution
- Hypertonic Solution
- Hypotonic Solution

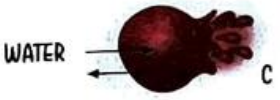
Animal Cell



Water entering the cell equals the water leaving it.

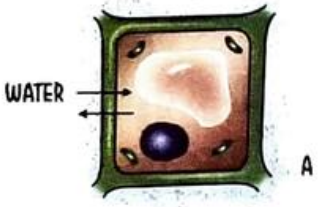


Water entering the cell is less than the water leaving it. (exosmosis occurs).

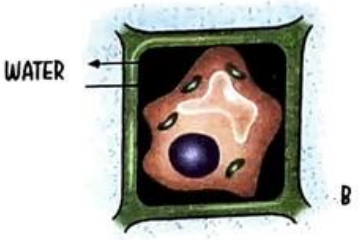


Water entering the cell exceeds the water leaving it. (endosmosis occurs).

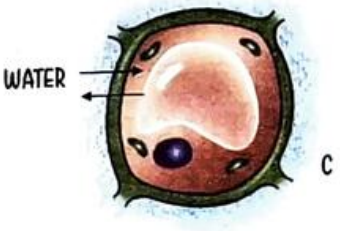
Plant Cell



No loss or gain of water by the cell



Water comes out of the cell. (exosmosis cocurs)



Water enters the cell. (endosmosis occurs)

TRANSPIRATION

CHAPTER: 4

Introduction

Loss of water in the form of vapours from the aerial parts of the plants through stomata, lenticels or cuticle is termed transpiration.

- Stomatal Transpiration
- Cuticular Transpiration
- Lenticular Transpiration

Significance of Transpiration

- Water and Nutrient Transport
- Temperature Regulation
- Gaseous Exchange
- Internal Water Balance
- Growth and Development
- Maintaining Turgor Pressure

GANONG POTOMETER

It is a device used to measure the rate of water uptake in a plant due to photosynthesis and transpiration under different conditions.

Factors affecting Transpiration

- Light
- Humidity in the air
- Temperature of air
- Velocity of wind
- Water availability

Adaptation in Plant to Reduce Transpiration

- Sunken Stomata
- Fewer Stomat
- Narrow Leaves
- Reduced Exposed Surface
- Loss of Leaves
- Thick Cuticle

GUTTATION

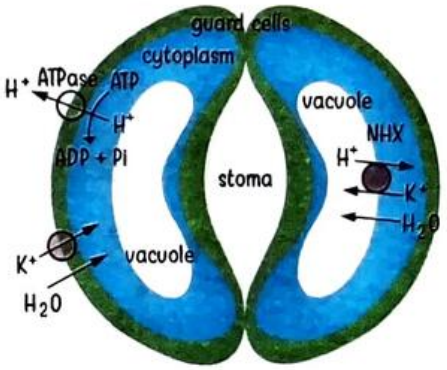
Guttation is the exudation of drops of xylem sap on the tips or edges of leaves of some vascular plants, such as grass, and a number of Fungi.

BLEEDING

Bleeding is the flow of the plant sap that occurs due to injury. The oozing liquid may contain sugars and amino acids.

The Potassium Ion Exchange Theory for Stomatal Transpiration

An increase in potassium (K) ion makes the guard cells hypertonic so that more water can enter inside the cells. During this, the stomatal pore is opened. Closing is just an opposite phenomenon of opening as in closing the water is removed from the guard cells and they become flaccid.



PHOTOSYNTHESIS

CHAPTER: 5

Introduction

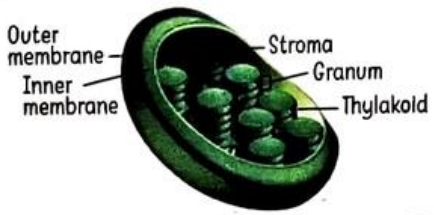
The process by which green plants and certain other organisms transform light energy into chemical energy by converting water, carbon dioxide, and minerals into oxygen and energy-rich organic compounds.

The general equation of photosynthesis is

$$\underset{\text{Carbon dioxide}}{6CO_2} + \underset{\text{Water}}{12H_2O} \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} \underset{\text{Sugars}}{C_6H_{12}O_6} + \underset{\text{Water}}{6H_2O} + \underset{\text{Oxygen}}{6O_2\uparrow}$$

CHLOROPHYLL: THE VITAL PLANT PIGMENT

- Found only in plant cells.
- Chloroplasts contain a green pigment called chlorophyll.
- Chlorophyll is essential for photosynthesis, the process by which plants make food.



STOMATA AND THEIR ROLE IN PHOTOSYNTHESIS

- Stomata are the minute openings occuring in large numbers on the lower surface of the leaves.
- The main function of the stomata is to let in CO_2 from the atmosphere for photosynthesis and when it is dark they close their openings to minimise loss of water through transpiration.
- Transpiration occurs along with photosynthesis and is the price the plant pays for photosynthesis.
- The stomata are opened and closed by the application of potassium ion concentration theorem.

Photosynthesis

Light reactions

Dark reactions

Photochemical phase

Light dependent reaction – Hill reaction

Biosynthetic phase

Light independent reaction – Calvin cycle

- In the light reaction, chlorophyll is activated, leading to the photolysis of water, release of oxygen, and the formation of ATP and NADPH through photophosphorylation.
- The dark reaction involves the combination of hydrogen, released by NADP, with CO_2 to form glucose.

Factors affecting Photosynthesis

External factors

- Light
- Water content
- Temperature
- CO_2 concentration

Internal factors

- Chlorophyll
- Structure of leaf
- Protoplasm

Factors Essential for Photosynthesis

- Chlorophyll
- Sunlight
- CO_2

PHOTOSYNTHESIS IMPORTANCE TO LIFE IN GENERAL

- Light Energy Conversion
- Glucose Synthesis
- Production of Oxygen
- Foundation of Food Chains
- Carbon Dioxide Reduction
- Habitat Formation

ADAPTATIONS IN PLANTS FOR PHOTOSYNTHESIS

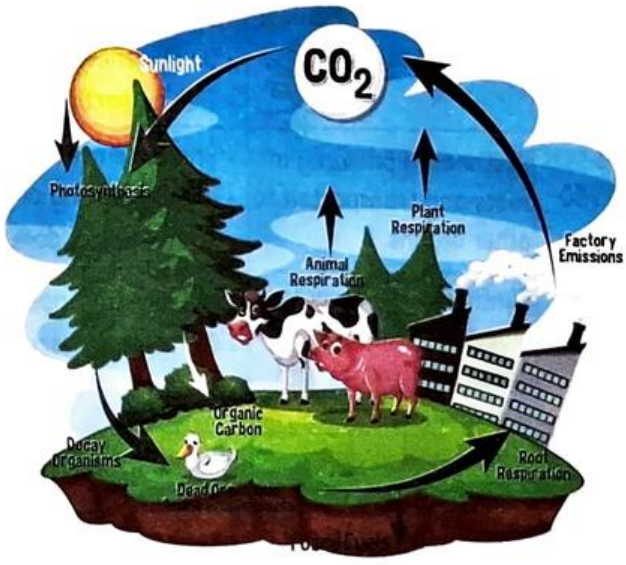
- Broad, Flat Leaves
- Chlorophyll Pigments
- Guard Cells Regulating Stomata
- Thin Cuticle on Upper Leaf Surface
- Palisade Mesophyll Cells
- Orientation Towards Light (Phototropism)
- Air Spaces in Spongy Mesophyll

CARBON CYCLE

The carbon cycle is a fundamental process for maintaining the balance of carbon in Earth's ecosystem, involving the movement of carbon among the atmosphere, oceans, soil, and living organisms. Human activities are currently disrupting this balance, leading to increased atmospheric CO_2 levels and climate change.

Essential Steps of Carbon Cycle:

(i) Photosynthesis
(ii) Food chains
(iii) Respiration
(iv) Decay
(v) Combustion
(vi) Heating limestone



CHEMICAL COORDINATION IN PLANTS

CHAPTER: 6

STIMULI AND RESPONSES

- The changes in the external or internal environment of an organism are known as stimuli and the resulting actions or movements caused by these stimuli are called responses.
- Though plants do not appear to have a nervous system or sense organs– their sensivity and co-ordination is the result of a **chemical control.**
- Plants respond to stimuli by producing some chemical compounds called **hormones** that work as messengers.

Plant Hormones

The term hormone was first used by William Bayliss and Ernest Starling in 1902.

Plant hormone	Discovered By	Site of synthesis	Functions
Auxins	F.W. Went in 1928	Synthesised primarily in shoot, apical meristems and young leaves.	• Promote cell elongation. • Suppress the growth of lateral buds. • Delay fall of leaves. • Induce formation of parthenocarpic fruits.
Gibberellins	Eiichi Kurosawa in 1926	Synthesised primarily in the meristems of apical buds and roots.	• Help in stem-elongation. • Break dormancy of seeds and buds. • Delay senescence. • Induce parthenocarpy.
Cytokinins	Skoog and Miller in 1950	Systhesised primarily in roots and transported to other organs.	• Stimulate cell-division and cell enlargement. • Prevent ageing of plant parts. • Inhibit apical dominance.
Ethylene	Dimitry Neljubow 1901	Synthesised in senescent leaves and flowers; germinating seeds and ripening fruits.	• Induces fruit ripening. • Promotes senescence.
Abscisic acid	Frederick T. Addicott 1960	Synthesised in green fruits and seeds at the onset of the winter period.	• Induces dormancy of buds and seeds. • Inhibits seed-germination and development. • Stimulates closing of stomata.

Trophic Movements in Plants

The biological phenomenon of growth or body movements of an organism in response to an environmental stimulus is known as tropism.

Phototropism	Movement in response to light.	**Eg.:** Sunflower 
Geotropism	Movement in response to earth's gravity.	**Eg.:** Plants Roots
Hydrotropism	Movement in response to water or moisture.	**Eg.:** Plant Roots
Thigmotropism	Movement in response to touch stimulus.	**Eg.:** Folding movement of ***Mimosa pudica*** leaflets 
Chemotropism	Movement in response to chemicals.	**Eg.:** Growth of pollen tubes during fertilisation

Movement	Root	Shoot
Phototropism	–ve	+ve
Geotropism	+ve	–ve
Hydrotropism	+ve	–ve
Thigmotropism	–ve	+ve
Chemotropism	+ve	ve

CIRCULATORY SYSTEM

CHAPTER: 7

Introduction

A system which transports various substances such as food, water, electrolytes, enzymes, hormones, antibodies and respiratory gases to or away from body tissues is called the circulatory system.

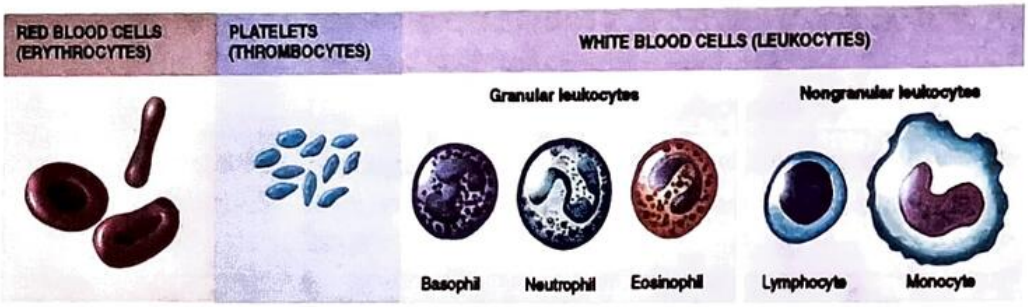
William Harvey (1628) was the first to discover the function of the heart and the circulation of blood.

MAIN COMPONENTS

Blood and lymph tissues, heart and blood vessels. (Arteries, veins, capillaries, lymphatic ducts, etc.).

Composition of Blood

- **Corpuscles (45%)**
  - R.B.C.
  - W.B.C.
    - Granulocytes
      - Neutrophils
      - Basophils
      - Eosinophils
    - Agranulocytes
      - Lymphocytes
      - Monocytes
  - Platelets
- **Plasma (55%)**
  - 90% Water
  - 10% Solid
    - Inorganic
      - Sodium Chloride
      - Bicarbonates
      - Calcium
      - Iron
      - Magnesium
    - Organic
      - Glucose
      - Amino Acid
      - Urea
      - Hormones
      - Enzymes



TISSUE FLUID AND LYMPH

Tissue fluid, also known as interstitial fluid, surrounds the cells of the body and provides them with nutrients and a means of waste removal.

Lymph is part of the lymphatic system and is a clear fluid that is similar to blood plasma. Its primary role is to return excess tissue fluid to the bloodstream and provide an avenue for immune responses, as it contains white blood cells, especially lymphocytes.

Blood-Coagulation

- Platelets $\xrightarrow{\text{Release}}$ Thromboplastin
- Prothrombin from Plasma $\xrightarrow[\text{Thromboplastin from platelets}]{Ca^{+2}}$ Thrombin
- Fibrinogen $\xrightarrow[+Ca^{+2}]{\text{Thrombin}}$ Fibrin
- Fibrin + Corpuscles $\longrightarrow$ Clot

Ca^{2+} is necessary for blood clotting.

RBC: DEFICIENT BUT MORE EFFICIENT

RBCs, though despite lacking certain cell components, are remarkably efficient in their primary function – transporting oxygen.

Lack of nucleus: increase the surface area-volume ratio for absorbing more oxygen.

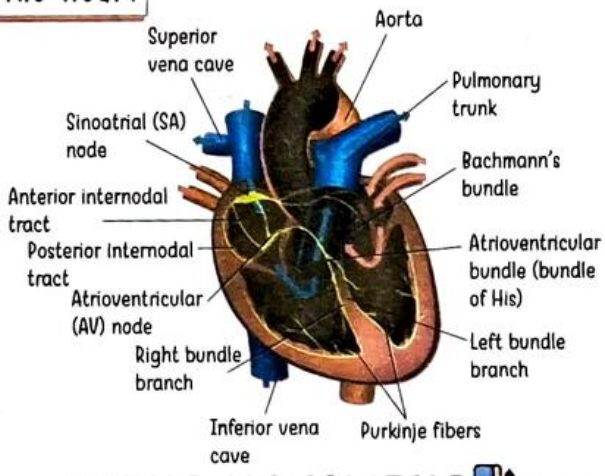
Lack of mitochondria: means fully transport of glucose in blood plasma, unused by RBC.

No endoplasmic reticulum: greater flexibility to navigate through tiny blood vessels.

Working of the Heart

The human heart has four chambers- two upper auricles and tow lower ventricles. The auricles have thinner walls as they receive blood from the body and pump it to the ventricles. The ventricles have thick muscular walls as they have to pump the blood to long distances.

Superior vena cava brings deoxygenated blood from the upper region of the body amd inferior vena cava brings deoxygenated blood from the lower region of the body to the heart. The pulmonary artery carries deoxygenated blood to the lungs for oxygenation. The aorta carries oxygenated blood to all parts of the body.



HEART BEAT (LUBB-DUP)

Refers to the sounds of the heart valves closing during the cardiac cycle. 'Lubb' is the first sound, occurring as the mitral and tricuspid valves close, and 'Dup' is the second sound, occurring as the aortic and pulmonic valves close.

SYSTOLE AND DIASTOLE

Systole refers to the phase when the heart muscles contract to pump blood out of the heart, while diastole is when the heart muscles relax, allowing the chambers to fill with blood.

The impulse or command which starts the heart beat, arises in Pace maker (Sino-atrial node, "SAN").

The main blood vessels entering and leaving the heart, the liver and the kidney

Organ	Blood Vessels Entering	Blood Vessels Leaving
Heart	Superior and Inferior Vena Cava (containing deoxygenated blood), Pulmonary Veins (containing oxygenated blood)	Pulmonary Artery (containing deoxygenated blood to lungs), Aorta (containing oxygenated blood to body)
Liver	Hepatic Artery (oxygenated blood from heart), Portal Vein (nutrient-rich blood from gastrointestinal tract)	Heapatic Veins (deoxygenated blood to inferior vena cava)
Kidney	Renal Arteries (carrying oxygenated blood from aorta)	Renal Veins (carrying deoxygenated blood to inferior vena cava)

BLOOD PRESSURE

The pressure exerted by blood on the walls of blood vessels is known as blood pressure.

To measure blood pressure, an instrument call a Sphygmomanometer is used.

- The standard blood pressure of a person is 120/80.
- The pulse rate of a person at rest is about 72 beats per minute.
- The duration of the cardiac cycle is 0·8 seconds.

EXCRETORY SYSTEM

CHAPTER: 8

Excretion is the removal of harmful and unwanted nitrogenous waste products from the body
The excretory system in human beings consists of: A pair of kidneys, a pair of Ureters, a Urinary bladder, and Urethra.

The excretory organs include kidneys, liver, skin, and lungs.

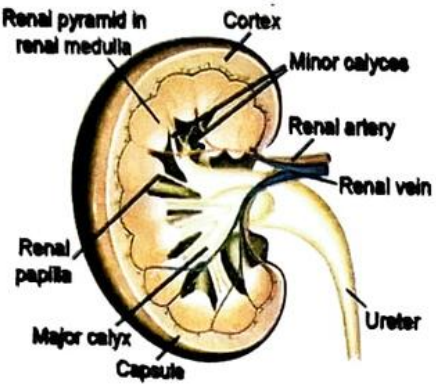
Excretory materials generated by some organs:

Organs	Excretory material generated
Kidney	Urine
Liver	Bile pigments bilirubin, biliverdin and urochrome are metabolic wastes of haemoglobin of dead red blood cells.
Skin	Sweat and metabolic wastes.
Lungs	Carbon dioxide and water vapour.
Large Intestine	Excess salts of calcium, magnesium, undigested food material and iron.
Lacrimal Gland	Small amount of nitrogenous wastes.

KIDNEY

The Kidneys are two, bean-shaped organ about 10 cm long and 6cm wide.
Kidney has two main regions: outer dark codex and inner lighter medulla.
Kidney plays a vital role in the maintenance of homeostasis such as:

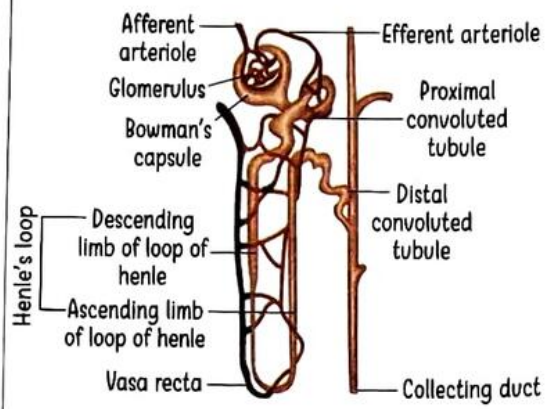
- Excretion of nitrogenous waste products.
- Regulation of water balance.
- Regulation of acid base balance.
- Elimination of other substance.
- Regulation of blood pressure.



NEPHRON

Kidney is composed of an enormous number of minute tubules called uriniferous tubules or nephrons or renal tubules. Nephroms are the structural as well as functional units of the kidney.
Functions of nephron:

- Filtration of blood.
- Returning useful susbtances to the blood so that they are not lost from the body.
- Removing substances from the blood that are not needed by the body.



MECHANISM OF URINE FORMATION

Urine is formed by the kidneys in the following three steps :

Ultrafiltration

Formation of a protein free filtrate from the plasma into the Bowman's capsule via the glomerlus.

Selective Reabsorption

Reabsorption of some useful filtered materials in the renal tubules.

Secretion

Collection of urine to the renal Pelvis and its passing out through the ureter into the urinary bladder and urethra.

Constituents of Urine

Organic in (g/L)		Inorganic in (g/L)	
Urea	2·3	Sodium chloride	9·0
Creatinine	1·5	Potassium chloride	2·5
Uric acid	0·7	Ammonia	0·6
Others	2·6	Others	2·5

Abnormal Constituents Found in Urine and the Diseases that it Indicates:

Components	Possible diseases
Excess sugar	Diabetes mellitus, glycosuria
Bile pigments	Jaundice/Liver cirrhosis
Blood cells	Haematuria/Kidney damage
Pus cells	Kidney infection/Pyuria
Albumin	High blood pressure, infection, injury or disease
Excess water	Diabetes insipidus (bland)

DISORDERS OF KIDNEY

Diuresis

Increased production of urine due to reduced secretion of ADH hormone.

Gout and kidney stones

Excess uric acid and certain salts like Ca^{+2} oxalate when get deposited in organs.

RENAL DIALYSIS

Renal dialysis is a device to treat kidney failure. This process uses an artificial kidney (hemodialyser) to remove waste and extra fluid from the blood. The blood is taken out from the body and filtered by this artificial kidney and than reintroduced into the body using a dialysis machine. Dialysis can be categorised into three primary types: in-center hemodialysis, home hemodialysis, and peritoneal dialysis.

OSMOREGULATION

The process by which osmotic concentration (amount of water, minerals and salt) of the blood inside the body of a living being is maintained, is known as osmoregulation.
The kidneys acting together with the hypothalamus and the pituitary gland are responsible for the osmoregulation of blood in the body of human beings.

Nervous System and Sense Organs (Eye and Ear)

CHAPTER: 9

The nervous system is a complex network of nerves and cells that carry messages to and from the brain and spinal cord to various parts of the body.

Nervous System

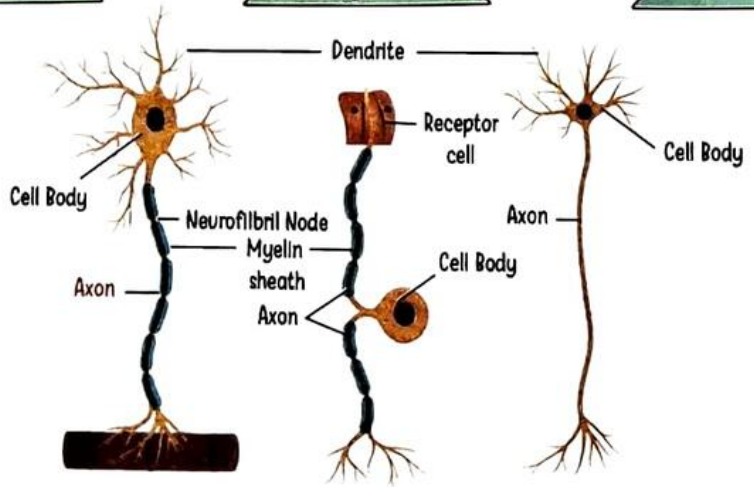
- **Central Nervous System (CNS)** (Site of information processing)
  - **Brain**
    - Cerebrum (consciousness intelligence, memory reasoning)
    - Cerebellum (body balance)
    - Medulla Oblongata (involuntary actions)
  - **Spinal Cord** (control and reflexes)
- **Peripheral Nervous System (PNS)** (Contains nerves that carry impulses to and from CNS)
  - **Somatic Nervous System**
    - Cranial Nerves (12 pairs)
    - Spinal Nerves (31 pairs)
  - **Autonomic Nervous System**
    - Sympathetic (Accelerates body sensation)
    - Parasympathetic (Restores normal body condition)

NEURON

It is the structural and functional unit of the nervous system. Each neuron consists a cell body with cytoplasm, nucleus, nucleolus, mitochondria, Nissl's granules, golgi structure and thin processes called dendrites, originating from dendrons (a relatively long extension) or axon (originates from the cell body).

Types of Neurons

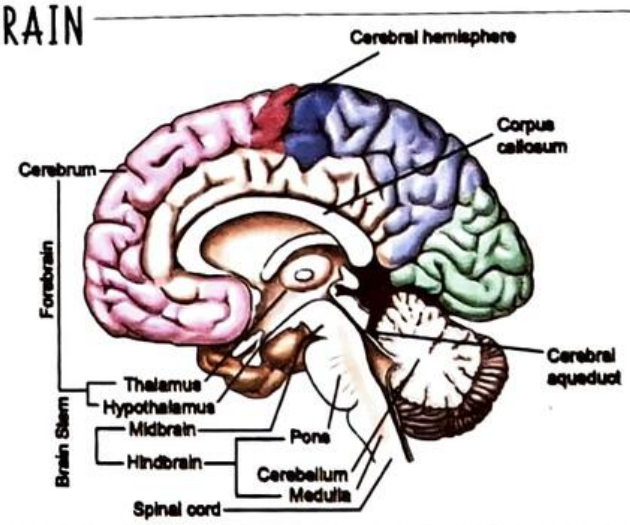
- Motor neurons
- Sensory neurons
- Interneurons



(i) Efferent (motor) neuron (ii) Afferent (sensory) neuron (iii) Interneuron

THE BRAIN

In proportion to the size of the body, the human brain is the largest among all animals. The brain is a very delicate organ, well- protected inside the brain box (cranium) of the skull.
The brain is protected by three meninges-outermost duramater, middle arachnoid and the inner vascular piamater. The cranial bones give extra protection to it.



Part of the brain	Function
1. • Fore Brain : Cerebrum • Hypothalamus, • Thalamus	• Intelligence, voluntary actions, memory, sensation, etc. • Control of body temperature, emotions , appetite, etc. • Relays pain and pressure.
2. Mid Brain : Optic nerves	• Vision and movement of eye balls and control of hearing by ears.
3. Hind Brain : Cerebellum Medulla Oblongata Pons	• Muscular co-ordination and balancing of body. • Involuntary actions of several internal organs. • It carries impluse from one hemisphere of the cerebellum to other.

Distribution of White and Gray matter in Brain and Spinal cord

The brain's gray matter, containing nerve cell bodies, is primarily found in the cortex and deeper structures like the basal ganglia, while white matter; composed of myelinated nerve fibres, forms connections between brain regions and between the brain and spinal cord.

The spinal cord shows a reverse arrangement: its gray matter is centrally located, forming an H-shape surrounded by white matter. This arrangement facilitates the processing and transmission of neural signals throughout the body.

VOLUNTARY AND INVOLUNTARY ACTIONS

Voluntary actions are those performed with conscious control, like deciding to read a book. Examples include playing a sport, writing, or cooking, where the individual is fully aware and in control of their actions. Involuntary actions, on the other hand, occur without conscious control, often as automatic responses. Examples include blinking, breathing, and reflex actions like sneezing.

SPINAL CORD

- The spinal cord transmits impulses from the periphery to the brain and vice–versa. It controls spinal reflexes routed along the reflex arc.
- There are 31 pairs of spinal nerves in humans, which emerge from the spinal cord at regular intervals as dorsal and ventral roots.
- There are 12 pairs of cranial nerves arising from the pons, medulla and the diencephalon. (between mid brain and cerebrum).

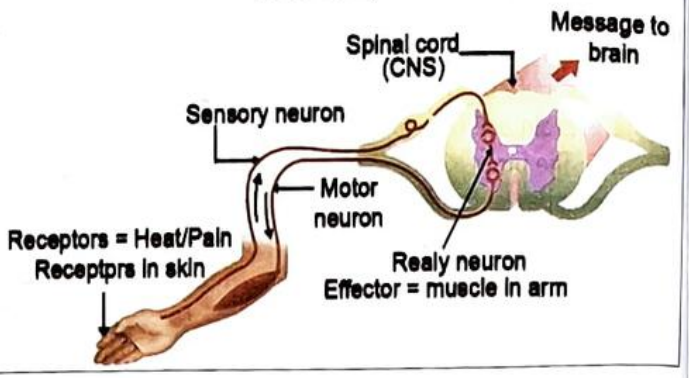
REFLEX ACTION

The involuntary, automatic, and often unconscious response to a stimulus is due to signals received by various nerve endings or receptors. The reflex arc is the pathway connecting a receptor to an effector organ.

Types of Reflexes

- Natural (inborn) Reflexes
- Conditioned (Aquired) Reflexes

The reflex arc, showing the pathway from receptor to effector:



Sense Organs

- The sense organs are the special structures which enable us to receive the stimulus and transmit it to the central nervous system.
- There are five senses in our body viz, touch, taste, smell, vision and hearing.

The Receptors for Sense Organs

Kinds of Receptors	
Light Receptors or Photoreceptors	The eyes are responsive to light and are capable of forming images.
Thermoreceptor Receptors	These receptors are of respond to changes in temperature rather than just heat and cold, as found, for e.g., in the skin.
Mechano-Receptors	These are sensitive to mechanical stimulations such as touch, pressure, pain and vibrations that producing sound.
Chemical Receptors	They can detect chemical changes in the environment which is a general property of protoplasm. Flavour of food, taste, smell etc. are detected by these receptors when vapours or liquid forms come into contact with these them.
Olfactory receptor	Receptor for smell, present in upper part of nasal chambers.

Eye

Sclera
- External layer (Collagen)
- Sense connective tissue
- Anterior portion-cornea

Choroid
- Middle layer
- Dense vascular layer
- Looks bluish

Retina
- Inner layer
- Three layered
- Inside to outside

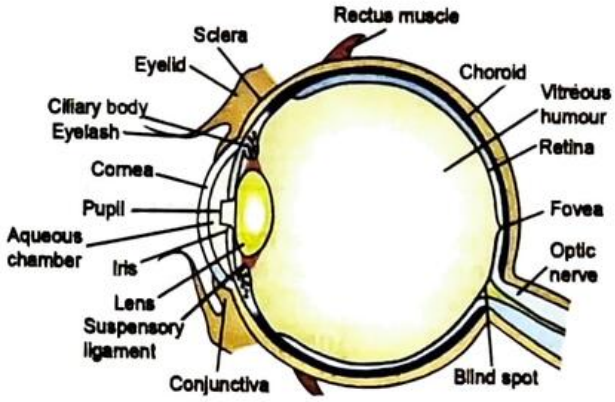
Ciliary body present at anterior part

Iris
- Pigmental and opaque structure
- Visible coloured portion of eye
- An aperture is surrounded by Iris called pupil

- **Photoreceptor cells**
  - **Rods**: Twilight or scotopic vision. Rhodopsin photopigment (Purplish-red protein visual purple). It is a derivative of Vitamin A
  - **Cones**: Day light vision (Photopic vision). Colour vision Different types of cones, photopigments respond to red, blue and green colour
- **Bipolar cells**
- **Ganglionic cells (inside)**

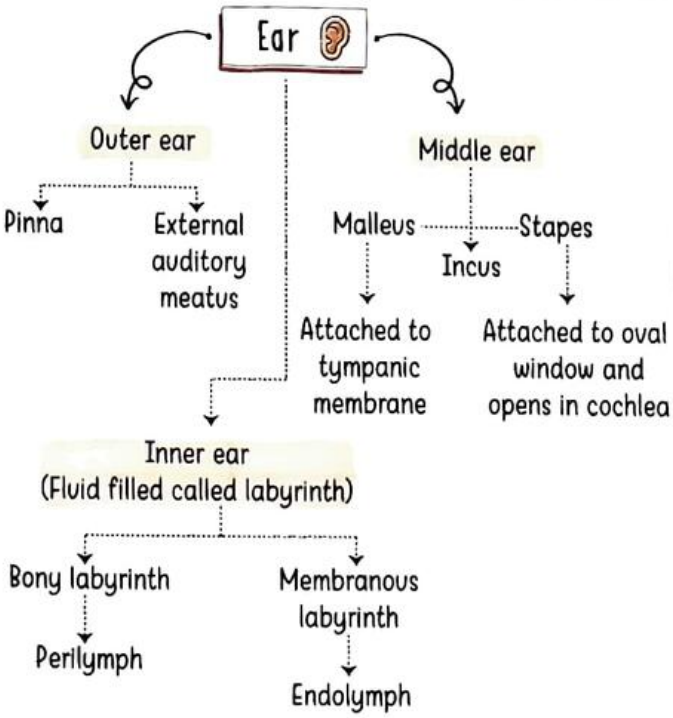
TERMS RELATED TO VISION

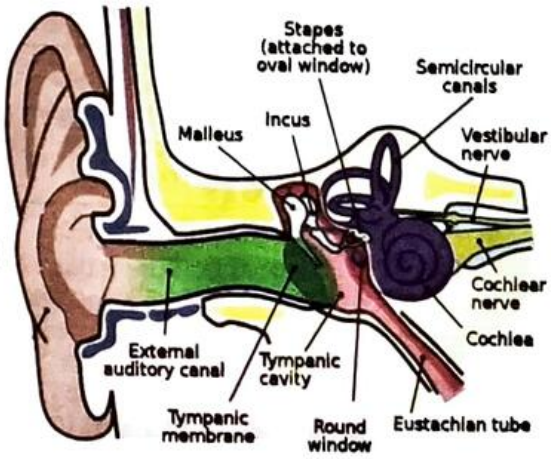
- **Stereoscopic Vision:** The ability of the eyes to perceive three-dimensional depth by combining two slightly different images from each eye.
- **Adaptation:** The eye's ability to adjust to varying light conditions, such as moving from darkness to light.
- **Accommodation:** The process by which the eye changes optical power to maintain a clear image or focus on an object as its distance varies



DEFECTS OF THE EYE

- **Myopia:** Nearsightedness; difficulty seeing distant objects clearly.
- **Hyperopia / Hypermetropia:** Farsightedness; difficulty seeing close objects clearly.
- **Presbyopia:** Age-related difficulty in focusing on close objects.
- **Astigmatism:** Distorted vision resulting from an irregularly curved cornea or lens.
- **Cataract:** Clouding of the eye's lens, leading to a decrease in vision.





Perception of Sound in the Human Ear: Sound waves enter the ear, are transformed into nerve signals by the cochlea, and then interpreted by the brain.

Role of the Ear in Maintaining Body Balance: The inner ear contains the vestibular system, semicircular canals, utriculus, and sacculus, which sense head movements and positions, sending signals to the brain to coordinate balance and spatial orientation.

ENDOCRINE SYSTEM

CHAPTER: 10

INTRODUCTION

Gland that secretes hormones into the blood stream in order to regulate life processes is known as endocrine gland. In humans, the main endocrine glands are the pituitary, thyroid, adrenal, pancreas and gonads.

Summary of Some Major Endocrine Glands

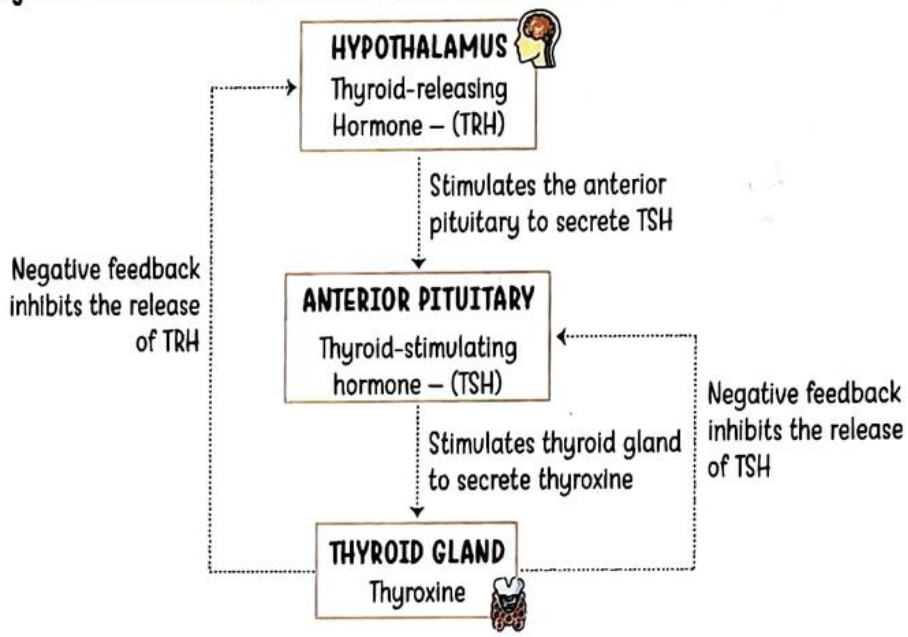
Endocrine Gland	Location	Hormones	Function	Disease/Condition due to Hyposecretion	Disease/Condition due to Hypersecretion
1. Pituitary (Master gland) (Pea shaped)	Attached to the hypothalamus of the brain.		Controls most other edocrine gland functions		
(a) Anterior pituitary (Adenohy-pophysis)	**Target organ:** Most tissue in the body	**(i) Growth hormone (GH)**	Stimulates general body growth.	**Dwarfism**	**Gigantism** in children, **Acromegaly** in adults.
	Target organ: Mammary gland	**(ii) Prolactin (PRL)**	Promotes lactation (milk formation and secretion)		
	Target organ: Adrenal cortex	**(iii) Adreno-corticotropic hormone (ACTH)**	Stimulates secretion from adrenal cortex		
	Target organ: Thyroid gland	**(iv) Thyroid-Stimulating Hormone (TSH)**	Stimulates thyroxine secretion		
	Target organ: Ovarian follicles in the female; Sertoli cells of seminiferous tubles in male	**(v) Follicle Stimulating Hormone (FSH)**	In females: growth of Graafian follicles, ovulation, and oestrogen secretion. In males: spermatogenesis.		
	Target organ: Ovary in females; Testis, Interstitial Leydig cells in males	**(vi) Luteinizing Hormone (LH)**	In females: maintenance of corpus luteum and secretion of progesterone, control the menstrual cycle. In males: testosterone secretion		Polycystic ovary syndrome (PCOS): cause of infertility and miscarriage in women

(b) Posterior (Neurohy-pophsis)	**Target organ:** Kidney 	**(i) Vasopressin (ADH- Anti-diuretic Hormone)**	Increases water absorption from kidney, contraction of blood vessels causing rise in blood pressure.	Diabetes insipidus 	Elevation of blood pressure
	Target organ: Uterus, mammary gland	**(ii) Oxytocin**	Causes contraction of the uterine muscle during child birth		
2. Thyroid Gland (Butterfly shaped) 	**Neck region,** surrounding the trachea	**(i) Thyroxine (T4), Triiodo-thyronine (T3)**	Regulation of metabolism, growth and differentiation	**Simple goitre:** enlargement of thyroid **Cretinism:** Dwarfism and mental retardation, **Myxoedema:** swelling on face and hands, sluggishness.	**Hyperthyroidism** (e.g., Graves' disease), **Exophthalmic goitre** (eye balls protrude, increased metabolism and restlessness)
		(ii) Calcitonin (Not in syllabus)	Promotes movement of calcium ions from blood to bones	Osteoporosis, or brittle bones 	
3. Parathyroid Gland (round or ovel-shaped (Not in syllabus)	**Behind the thyroid gland** 	**Parathyriod Hormone (PTH)**	Regulation of calcium and phosphate levels in blood	**Hypocalcemia,** Tetany	**Hypercalcemia:** Kidney stones, bone pain and muscle weakness.
4. Adrenal Gland (Triangular/ pyramidal shaped) 	**On top of each kidney** **(i) Adrenal Cortex:**	(a) Mineralo-corticoid (aldosterone), (b) Glucocor-ticoids (cortisol corticosterone), (c) Sex corticoids (adrenal androgens)	Salt-ion balance, regulation of metabolism Stimulates the development of external sex charact-eristics in males	**Addison's disease (Cortex),**	**Cushing's syndrome** (due to excess cortisol), **Adrenal virilism**
	(ii) Medulla	**Adrenaline (Epinephrine), Noradrenaline (Norepineph-rine)**	Causes stimulation of nervous system, **response to stress, prepares body for any emergency**		Prolonged fight of flight symptoms which wears down the individual.
5. Pancreas (Elongated, leaf-like shaped) 	Behind the stomach, **from Islets of Langerhans** 	**Glucagon (Alpha cells),** **Insulin (Beta cells)**	Regulation of blood sugar level  Raise blood sugar level Lower blood sugar level	Diabetes Mellitus (Hyperglycemia type 1 (due to insulin deficiency)	Insulin shock (Hypoglycemia)

		Somatotropin (Delta cells)- (not in syllabus)	inhibit secretion of glucagon and insulin.		
6. **Gonads** (Not in syllabus) (i) **Ovaries (in femel)** (almond-shaped)	**Pelvic cavity**	**Oestrogen, Progesterone**	Regulation of female reproductive system, maintenance of secondary sexual characteristics	Infertility, Menstrual irregularities	
7. **Testes (in males)** (Oval-shaped)	**Scrotum**	**Testosterone**	Regulation of male reproductive system, maintenance of secondary sexual characteristics	Hypogonadism, infertility	Androgen excess, Precocious puberty

Hormone: Its a secretion from some glandular part of the body, which is poured directly into blood and which acts on the target organs.

Negative Feedback Mechanism with Refrence to the TSH Level in Blood



When the level of thyroxine concentration exceeds the normal level in blood, then it inhibits the secretion of TRH by hypothalamus and the secretion of TSH from the anterior pituitary gland.

THE REPRODUCTIVE SYSTEM

CHAPTER: 11

Reproductive system involves the organs and structures that take part in offspring production (fertilisation).

Reproduction

Biological process by which new individual organisms are produced from their parents.

- Asexual (Single parent is involved)
- Sexual (Both the parents are involved)

REPRODUCTIVE ORGANS

Male Reproductive System

Testes (one pair)

Produce sperms and hormones

- Seminiferous tubule (Produces sperms)
- Interstitial cells (Leydig cells)

Produce testesterons

Ducts

Stores or transports the sperm to the exterior

- Epididymis
- Vas deferens
- Ejaculatory duct

Accessory glands

Add secretions to the sperm, constituting part of the semen

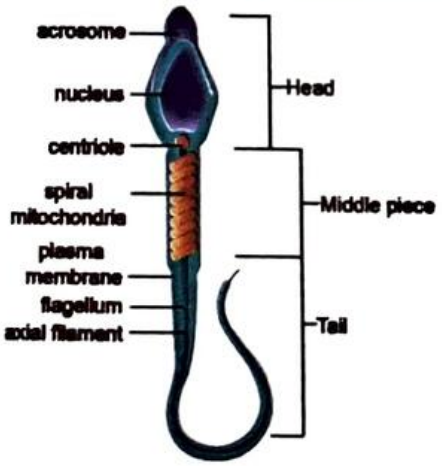
- Seminal vesicles
- Prostate gland
- Cowper's gland (Bulbo-urethal gland)

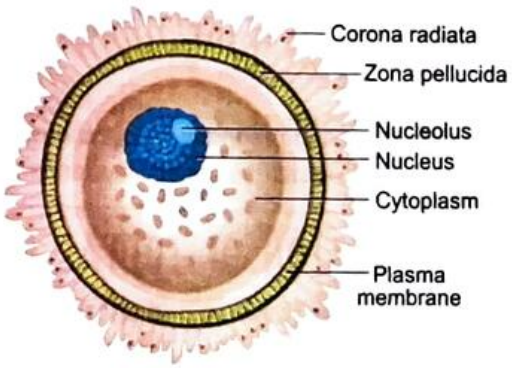
Supportive Structure-Penis

- Transfering the sperm into the female vagina
- Common passage for sperms and urine

Spermatogenesis : The process of transformation of male germ cells, spermatogonia, into spermatozoa is called spermatogenesis.

Structure of a Sperm and an Ovum





Female Reproductive System

- **Ovaries (One pair)** → Produce ova and hormones
- **Oviduct (One pair)** → Transports ova to uterus
- **Uterus** → Protects and nourishes the developing embryo
- **Vagina** → Receives the sperms
- **External organs** → Vulva

Oogensis: It is the process in which the ova producing cells are rise to the mature ovum.

Secondary Sexual Characters in Males and Females

Male	Female
Growth of facial, pubic, and body hair	Growth of pubic and underarm hair
Deepening of the voice as the larynx enlarges	Breast development
Development of larger Adam's apple	Widening of the hips
Increased muscle mass and strength	Menstruation begins (menarche)
Growth of the testes and penis	Increase in size of uterus and vagina

FERTILISATION

The process of union male (sperm) and female (ovum) to form a zygote gametes is called fertilisation.

GESTATION PERIOD

The period from conception to birth, i.e., the full term of development of the embryo in the uterus. In humans, it is around for 270 days.

IMPLANTATION

The fixing of the blastocyst to the wall of uterus/endometrium.

PARTURITION

It is the act of expelling the full term foetus from the mother's uterus at the end of gestation.

OVULATION

It is the release of mature ovum or oocyte from the ovary, at the 12 day of menstrual cycle.

PUBERTY

It is the achievement of reproductive maturity in humans. The males attain puberty at the age of 13 to 14 years, while females attain it at the age of 10-12 years.

PLACENTA

An organ, formed in pregnant female that connects the developing foetus to the uterine wall, allowing nutrient uptake, waste elimination, and gas exchange through the mother's blood supply.

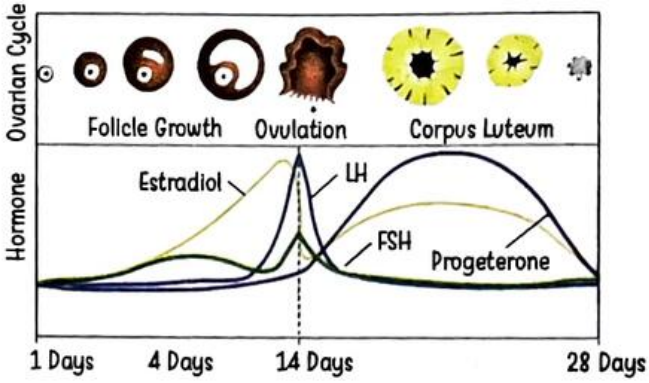
FUNCTION OF FOETAL MEMBRANES AND AMNIOTIC FLUID

Foetal membranes (amnion and chorion): Enclose and protect the foetus against physical shocks,; amnion produces amniotic fluid; chorion forms part of the placenta and contributes to the transfer of nutrients from mother to foetus.

Amniotic fluid: Cushions and protects the foetus; maintains constant temperature; allows foetal movement; helps in foetal development.

MENSTRUAL CYCLE

It is a monthly process of ovulation and menstruation in females, preparing the body for pregnancy.



MENARCHE

It is the first menses that occurs and it signals the onset of puberty in females at about the age 11 to 14

MENOPAUSE

It is the cessation of menstrual cycle in female at around the age of 45 to 50 years.

ROLE OF SEX HORMONES IN REPRODUCTION

The main reproductive hormones oestrogen, testosterone, and progesterone are instrumental in sexuality and fertility. They are responsible for pregnancy, puberty, menstruation, menopause, sex drive, sperm production and more. These hormones are produced in the ovaries (in females) and testes (in males).

Stage in the Development of Human Embryo

- (from ovary)
  - Egg (ovum)
  - Unfertilised egg
  - released from ovary
- In Oviduct
  - Zygote
    - Fertilised egg (1-cell stage)
  - Morula
    - Spherical mass of cells, resulting from repetative division of zygote
    - (8-16 blastomeres)
  - Blastocyst
    - Hollow sphere of cells
    - Trophoblast and inner cell mass formation
- Inside Uterus
  - Embryo
    - Size of a large pea, hardly resembles with human-being
  - Advance Embryo
    - (Heart and blood vessels formed (5 weeks)
  - Foetus
    - Limbs appeared (resembles with human being)
  - Infant
    - Born at the end of gestation period

Twins and its types

Identical Twins

These are monozygotic twins. They are formed from a single zygote or fertilised egg. The zygote splits into two parts and each part develops into an individual embryo. Their genetic origin is the same and therefore they have the same DNA.

Fraternal Twins

These are dizygotic twins. They are formed from two eggs which are fertilised by two different sperms. They share 50% of their genetic traits, just like any other sibling born at different times.

POPULATION: THE INCREASING NUMBERS AND RISING PROBLEMS

CHAPTER: 12

POPULATION

It is defined as the total number of individuals of a species present in a geographically localised area.

POPULATION EXPLOSION

The rapid increase in population is termed as population explosion.

DEMOGRAPHY

The statical study of human population is termed as demography.

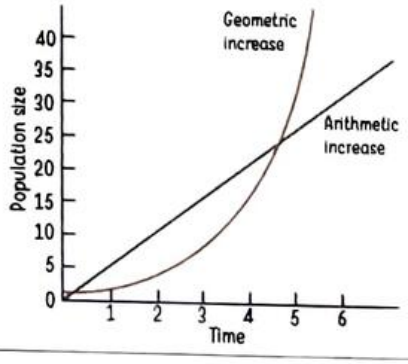
POPULATION DENSITY

It is defined as the size of the population in relation to unit area at given instant of time.
D = N/S, where D = density of population,
N = number of individuals, S = unit of space.

DYNAMICS OF POPULATION GROWTH

Exponential Growth: Growth at a increase per unit time. (Geometric)
Arithmetic/Logistic Growth: Growth at a constant amount per unit time

Population increase in geometrical (exponential progression – 1, 2, 4, 8, 16....etc., whereas food production rises in arithmetic progression- 1, 2, 3, 4, 5,.....etc.



Factors Responsible for Population Explosion in India

- Poverty and Illiteracy
- Agricultral advancement
- Medical facilities
- Religious and social customs
- Desire for a male child
- Lack of awareness
- Decrease in infant mortality

Problems Arisen due to Population Explosion

- Unemployment
- Over exploitation of natural resources
  - Forest
  - Energy
  - Water
  - Mineral
- Low per capital income
- Unexpected price rise
- Pollution
- Pressure on Infrastructure
- Inadequate food

CENSUS

It is the counting of existing population at a fixed time as regards to name, age, sex, occupation, date of birth, matrimonial status, number of dependents in a family, race, religion, language, education, residence etc. In India, a census is carried once in every 10 years.

- Birth Rate (Natality) = $\frac{\text{Number of live births in a year}}{\text{Population in the middle of that year}} \times 1000$
- Death Rate (Mortality) = $\frac{\text{Number of deaths in a year}}{\text{Total population in the middle of the year}} \times 1000$
- Fertility Rate = $\frac{\text{Number of live births in a year}}{\text{Number of women aged 15 to 45 in the year}} \times 1000$
- Infant Mortality Rate = $\frac{\text{Number of babies less than 1 year old in a given year}}{\text{Number of babies born in that year}} \times 1000$

- **Growth rate of population:** It is calculated as difference between birth rate and death rate, change in population size for a particular region over a given time period.

Methods of Population Control

- Family planning
  - Family Welfare Society
  - Inverted Red Triangle is the symbol of this socity
- Awareness or education
- Birth control techniques
  - Non-surgical methods
    - Contraceptive pills
    - Condom
    - Diaphragms
    - IUDs
  - Surgical methods
    - **Tubectomy** — Fallopian tubes are cut and tied up in femals
    - **Vasectomy** — Cutting and tying the vas-deferens in males
- Strictly enforce to increase marriage age

MEDICAL TERMINATION OF PREGNACY (MTP)

It is a legalised method of termination of pregnancy. It is also known as induced abortion to get rid of unwanted pregnancies due to (i) casual unprotected intercourse, (ii) failure of the contraceptive used during coitus, and (ii) rape. It is also essential in cases where continuation of pregnancy could pose threat to the life of either mother or foetus or both.

SEX RATIO

The sex ratio (or gender ratio) is the ratio of females to males in a population. The ratio tends to be 1:1 in most sexually reproducing species, which is explained by Fisher's principle. Many species deviate from an even sex ratio, either periodically or permanently.

HUMAN EVOLUTION

CHAPTER: 13

EVOLUTION

Evolution is a slow and continous process whereby complex from of life have emerged from simple forms through million of years.

Evidences of Evolution

Homologous Structures
Same structure, different function; e.g. wings of bat, flipper of whale, arm of human.

Fossils
Provide a visual record of how organisms changed over time.

Embryology
Similarities in the early development of vertebrate embryos; e.g. early embryonic development of humans and monkeys.

Analogous Structures
Unrelated organisms, similar physical structures; e.g. wings of bat, wings of bird, wings of butterfly.

Vestigial Structures
Structures which once had a function, but no longer do; e.g. vestigial hind limbs of modern whales.

Molecular Biology
Number of difference in amino acid sequence; e.g. sequence of human haemoglobin identical to chimpanzee while less closely related primates have greater differences.

Autogenesis Theories for Origin of Life

Theory of Inheritance of acquired charactor

Given by French biologist Jeam Beptiste de Larmarck (1744-1829)

On the basis of fossils and matching the organism with their environment

Use and Disuse of Organs

Eg: neck and forelimb of Giraffe

Eg: Vestiginal organ [Wisdom teeth vermiform appendix Ear pinna - in humans]

Lamarck prostulated that the organism could pass its modifications to the offsring after generations.

Theory of Natural Selection

Given by English Scientist Charles Darwin (1809-1882)

On the basis of observations, Put some strong conclasions

(i) Rate of reproduction in species
(ii) Struggle for existence
(iii) Variations
(iv) Survival of fittest

Concluded that the fittest organism survive and reproduce, passing on their advantageous traits to the next generation, leading to the evolution of species over time.
Eg: Peppered moth adaptation.

VESTIGINAL ORGANS

Organs that have ceased to be any use to the prossessor but still persist generation in reduced form.



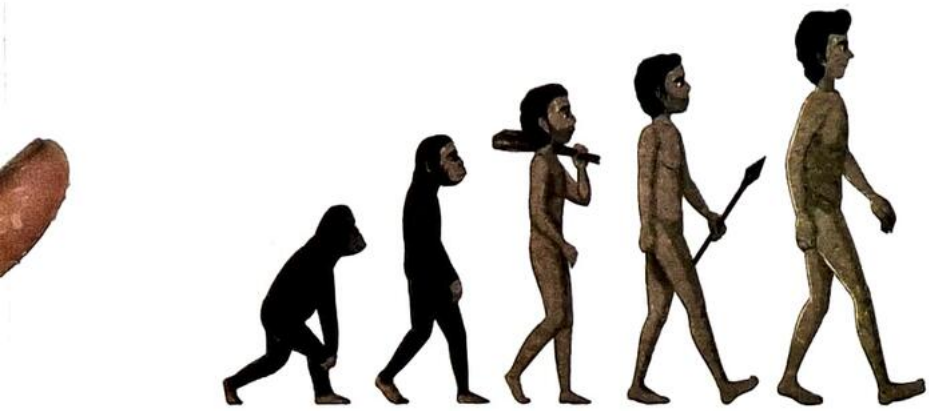
PEPPERED MOTH (BISTON BITULARIA)

During the Industrial revolution in Britain, dark colour moths become more common than the ligher ones because they were less visible to predators, prove the theory of natural selection.



Different stages of human evolution and their major characteristics

	Australopithecus	Homo Habilis	Homo Eractus	Neanderthal	Cro-Magnon Man	Modern man/ *Home sapiens*
Bipedalism	Walked nearly straight; vertebral column had a definite lumbar curve.	Bipedal gait; slightly more erect than Australopithecus	Walked fully eract	Absolute bipedalism	Perfect bipedalism	Perfect bipedal locomotion
Cranial capacity	450 to 600 cm^3	680 to 735 cm^3	800 to 1125 cm^3	about 1450 cm^3	1450-1600 cm^3	1450-1600 cm^3
Size of canine teeth	Almost the same as modern man; did not project above the level of other teeth	Small canines; almost like that of modern man.	Canines of lower jaw larger more like that of modern man	Small canines; heavier than that of modern man; presence of wisdom teeth	Small canines; teeth closer to each other; wisdom teeth present	Small canines; teeth closer to each other; wisdom teeth present
Forehead and brow ridges	Low forehead; eyebrow ridges projecting over the eyes	Forehead receding with ape-like head	Forehead receding and brow ridges; deep set orbits; proganathous face	Forehead low and slanting broad and flat; prominent eye brow ridges	Broad arched forehead; moderate brow ridges; orthognathous face	Steep forehead; reduced brow ridges; flattened face
Development of chin	Lack of chin; prognathous face	Almost no chin	Almost no chin	Strong lower jaw with almost no chin	Well developed chin	Prominent chin; snout disappeared
Body hair	Body covered with hair	Hair on body reduced	Hair on body reduced	Reduced hair on body	Body less hairy	Highly reduced body hair
Height and posture	120 cm (3.5 to 4 ft); nearly upright	120 to 150 cm (4.5 to 5 ft); bent kneed posture	120 to 150 cm (5 to 5.5 ft); straight legged stance or upright	160cm (about 5.5 ft), semi-erect stooping posture	180 cm (5.5 cm to 6 ft); fully erect	Upto 190 cm; fully erect



POLLUTION: AN INCREASING ENVIRONMENTAL HAZARD

CHAPTER: 14

INTRODUCTON

Pollution can be defined as "the undesirable changes in the physical, chemical or biological characteristics of air, water and land that are harmful for health, survival or activities of human or other living organisms."

Types of Pollution

- **Air Pollution**: Vehicular, Industrial, Burning Garbage, Brick kilns
- **Water Pollution**: House hold detergent, Sewage, Industrial waste, Oil spills
- **Soil Pollution**: Industrial, Urban, Commercial, and domestic chemical fertiliser
- **Thermal Pollution**: Power plants, oil refineries, Nuclear power plant
- **Biomedical Waste**: Needles, syringes dressings etc.
- **Radiation**: X-rays, UV rays, radioactive fallout, uranium mining
- **Noise**: Vehicles, Industrial, construction sites, flight zones, loudspeakers

Categories of Wastes

Biodegradable

Waste material which can be broken down or decomposed in nature into non-poisonous substance by the action of microorganism like bacteria, fungi etc.

Eg: Fruits and vegetables, peels, flowers, plant, paper, wood, kitchen wastes.

Non-biodegradable

Waste material which can not be broken down into non-poisonous substance in nature.

Eg: Plastic, polythene beg, glass, styrofoam, synthetic fibre, silver foil, pestilides like DDT etc.

Effect of Pollution

- Climate change
- Environment hazard
- Human, animal and plants health
- Agriculture hazard
- Ecosystem imbalance
- Disrupted biogeochemical cycle

ACID RAIN

Acid rain is rainfall made acidic when atmospheric water combines with atmospheric pollution like industrial burning of coal and other fossil fuels, the waste gases which contain sulphur and nitrogen oxides.

It causes environmental harm, buildings damage, wildlife harm, and disrupts ecosystems.

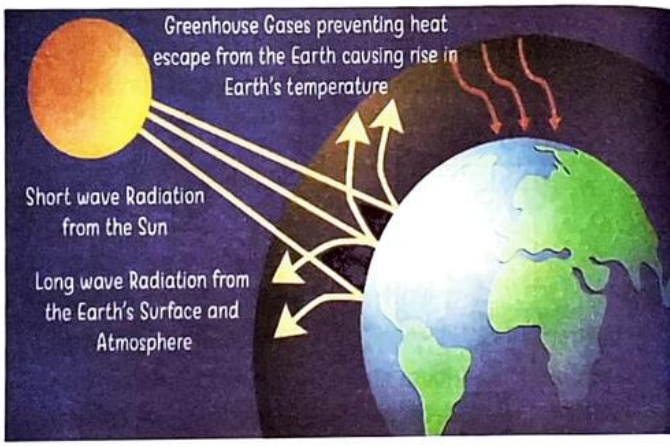
OZONE LAYER DEPLETION

Caused by the release of chlorofluorocarbons (CFCs) and other ozone-depleting substances (ODS) which break down ozone (O_3) molecules in the stratosphere, leads to increased UV radiation reaching the Earth, causing health hazard like skin cancer, cataracts, and harm to ecosystems.

GREENHOUSE EFFECT AND GLOBAL WARMING

This process occurs as a result of increased emission of CO_2, CH_4, hydrofluorocarbons, chlorofluorocarbons (CFCs) and water vapour etc. by man-made or anthropogenic causes start accumulating in Earth's atmosphere and prevent to escape heat, thus warming the air.

Global warming is already affecting living beings by such as increased ocean levels, droughts and changing weather patterns.



Measures to Control Pollution

- Use of unleaded petrol/CNG in automobiles
- Installation of tall chimneys, filters in factories
- Switch off engines at traffic signals
- Social forestry
- Setting of sewage treatment plant
- Ban on polythene and plastic
- Organic farming
- Applying 3 R's Reduce Reuse Recycle

EURO BHARAT VEHICULAR STANDARD

To significantly reduce the permissible emission levels of pollutants from vehicles, since 2000, Euro norms are followed in India under the name Bharat Stage Emission Standards, instituted by the Central Pollution Control Board (CPCB), within the Ministry of Environment Forests and Climate Change. The latest standard in effect across India is BS VI.

SWACHH BHARAT ABHIYAN

Swachh Bharat Abhiyan (SBA) or Clean India Mission is a nationwide campaign, aimed at cleanliness and sanitation, initiated by the Government of India on 2 October 2014. It seeks to eliminate open defecation and improve solid waste management. Key features include constructing toilets, promoting hygiene, and educating citizens about the benefits of cleanliness for overall well-being.

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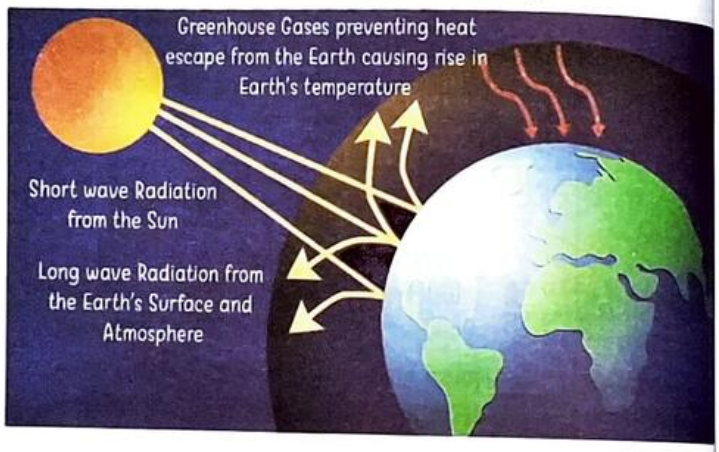
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